

Short communication

Intelligent resource allocation in hybrid RF/LiFi networks via deep deterministic policy gradient based DRL mechanism

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ABSTRACT

A hybrid radio frequency (RF) and light fidelity (LiFi) network combines the strengths of RF and LiFi technologies. RF offers broad coverage, while LiFi provides high data rates. As these technologies operate on non-interfering spectra, they can co-exist without interfering with each other. This setup not only boosts data rate but also makes the network more reliable, especially when physical obstacles might block signals. However, resource management in hybrid RF/LiFi networks is challenging because of the dynamic environment and the different characteristics of the two technologies. Efficient resource allocation maximizes the data rate in these networks. In this paper, we introduce a model-free deep reinforcement learning (DRL) approach to solve the resource allocation problem in hybrid RF/LiFi networks. Our DRL model is designed to handle real-world conditions, considering factors like blockages and user mobility. Unlike traditional methods that need extensive modeling and assumptions, our approach learns directly from interacting with the environment, making it highly adaptable and robust. Through simulations, it is observed that our method improves resource utilization and overall network performance, achieving a 62.8% increase in sum rate and a 42.8% improvement in optimal transmit power compared to conventional methods.

1. Introduction

In the recent years, wireless internet data traffic has seen a phenomenal growth across the radio frequency (RF) spectrum [1]. The growth is estimated to fully saturate the RF spectrum by 2035 [2], highlighting significant future challenges. Optical wireless communications (OWC), such as light fidelity (LiFi), present a promising possibility to alleviate the burden on RF communications. LiFi uses the unlicensed light spectrum, offering high-data rate and bidirectional, and multi-user connectivity without interfering with RF signals [3–5]. However, challenges such as inefficiency in non-line-of-sight (NLOS) scenarios, random orientations, and user equipment (UE) mobility introduce significant channel quality variations, underscoring the need to integrate RF and LiFi communications for robust and mobile broadband solutions [6,7]. The integrated system, known as a hybrid RF/LiFi system, has emerged as a valuable addition to heterogeneous networks (HetNets), alleviating the strain on overloaded only-RF communication systems [8].

In hybrid RF/LiFi systems, optimal resource allocation remains a crucial inherently complex integer non-concave problem. Conventional

convex optimization based resource allocation algorithms often fail in obtaining global optimum solutions to such problems. These solutions are based on presumption of values for at least one parameter. However, presuming a value does not make the system robust for optimal resource allocation. Deep Q-Networks (DQN) based mechanisms can overcome this issue. However, it's limitation lies in its capability to manage only discrete and low-dimensional action spaces and fails in dynamic systems with large dimension and continuous action spaces [9–11]. Therefore, we are motivated on exploring a machine learning technique designed to optimize future rewards in a large dynamic environment. Our objective is to devise an intelligent system that maximizes the overall average throughput for all UEs. To address these objectives, we will apply off-policy deep reinforcement learning (DRL), which is a fusion of reinforcement learning and deep learning techniques [12].

In this paper, we simulate an environment where both RF and LiFi-enabled APs are present in a room, as depicted in Fig. 1. We propose a model-free off-policy actor-critic algorithm to learn policies in high dimensional continuous action spaces [13]. Deep deterministic

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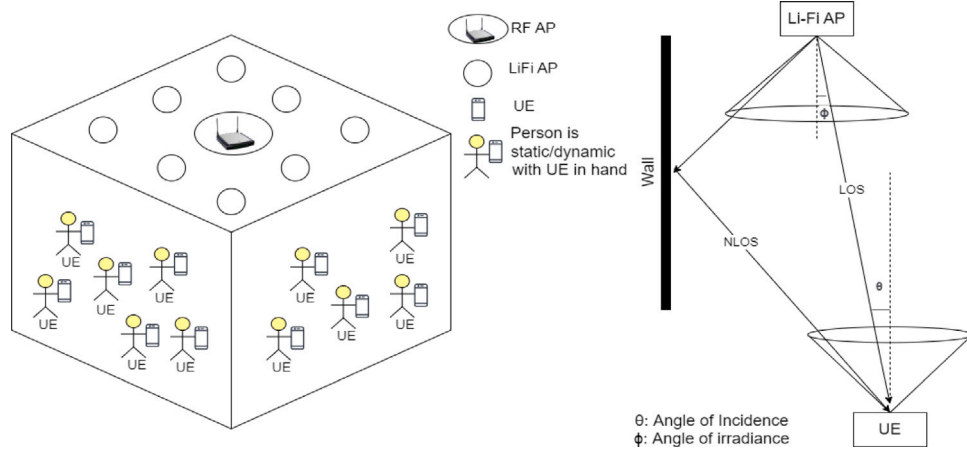


Fig. 1. Hybrid RF/LiFi environment.

policy gradient (DDPG) is a off-policy-based DRL technique, where an agent can pick a policy not solely from it's current policy (observational policy) but also from it's past experiences (behavioral policy). A key feature is that it is simple and easy to implement. Our main contribution in this paper are as follows

- **Novelty:** We have developed a virtual environment for a hybrid RF/LiFi system that closely emulates real-world conditions, including their dynamicity. To the best of our knowledge, this is the first instance of such a system being created with the application of Deep deterministic policy gradient (DDPG).
- **Dynamic set-up:** Our model accounts for the dynamic nature of the environment by considering the introduction of new UEs into the room and incorporating load balancing constraints, thus reflecting a realistic and dynamic system configuration.
- **Continuous action space:** The proposed state-of-the-art algorithm utilizes a continuous state and action space, resulting in more optimal resource utilization, higher achievable data rates, and reduced transmission power.
- **DRL algorithm:** Using the DRL based DDPG algorithm which can handle continuous state and action space. A comprehensive framework has been made to implement the optimization goal of maximizing the achievable sum-rate.

The rest of the paper is organized as follows: Section 2 explains system model under consideration. Section 3 explains the problem considered and Section 4 represents optimization goal to solve the problem formed. Section 5 describes the illustrated framework. In Section 6, simulations have been explained and Section 8 concludes the paper.

2. System model

Fig. 1 illustrates an indoor scenario, typically representing a hall where users may be static or moving at variable speeds, i.e., they are dynamic. Each user holds a UE. Multiple light-emitting diode (LED) access points (APs) are deployed on the ceiling, known as LiFi APs. Additionally, a RF AP is installed on the ceiling, creating a hybrid RF/LiFi setup. Together, the RF and LiFi APs provide uninterrupted connectivity to the UEs. In this indoor scenario, UEs connect to the LiFi AP if they are within the line-of-sight (LOS). When LOS components are absent from the LiFi AP, UEs connect to the RF AP to maintain ubiquitous connectivity.

The another diagram showing an enlarged version of the light communication from the LiFi AP, providing significant details about the LiFi communication model. This link diagram illustrates the relationship between the angle of irradiance from the LiFi AP coverage area and the angle of incidence at the UE's field-of-view (FOV).

The link diagram demonstrates communication between the LiFi AP and a UE, illustrating both the direct LOS and indirect paths (NLOS) through which light signals can travel from the LiFi AP to the UE in a hybrid RF/LiFi system. The LOS path provides the strongest signal. In contrast, the NLOS path, involving reflections, may result in weaker and more distorted signals due to delays and scattering.

The light emitted by the LiFi AP forms an angle ϕ , which represents the angle of irradiance. This angle is between the vertical axis (normal to the LiFi AP surface) and the direction in which light is emitted. When received at the UE, the light forms an angle θ , representing the angle of incidence. This angle is between the vertical axis (normal to the UE surface) and the direction from which the light is received. Both angles are crucial as they affect the coverage area, signal strength, and overall system performance.

3. Problem formulation

Fig. 1 depicts the system under consideration. The set $\mathcal{E}$ denotes the AP ensemble, with individual APs indexed as $e = 0, 1, 2, \dots, |\mathcal{E}|$. The RF AP is situated at the center of the ceiling and is indexed as $e = 0$. The LiFi APs, serving as LED light sources, are indexed as $e = 1, 2, \dots, |\mathcal{E}| - 1$. The set $\mathcal{F}$ comprises UEs with indices $f = 0, 1, 2, \dots, |\mathcal{F}|$. Each UE is assumed to maintain a constant height t above the floor. UEs have the option to establish connections with either LiFi or RF APs to receive data, contingent upon the AP offering superior data rates. Additionally, the model accommodates the inclusion of a new UE entering the scenario.

3.1. Parameters

To explain the formulation of the problem with clarity, we first explain the parameters involved. The problem formulation involves the following parameters

3.1.1. LiFi channel gain

Let a UE f connect to a LiFi AP in LOS. The direct current (DC) channel gain is governed by the Lambertian law model as [14]

$$G_{ef}^v = \frac{(m+1)A_{pd}\cos^m\theta_{ef}\cos\phi_{ef}T_{opt}(\phi_{ef})g(\phi_{ef})}{2\pi d_{ef}^2}, \quad (1)$$

where $(T_{opt}(\phi_{ef}))$ is the receiving optical filter's gain, which is either 1 or a constant value within the receiver's field of view (FOV), θ_{ef} denotes the incidence angle at UE f from the e th AP, ϕ_{ef} is the angle of irradiance at AP e , d_{ef} represents the distance between the f th UE and the e th AP, and m represents the exponent of the Lambertian radiation pattern, expressed as $m = -\frac{\ln 2}{\ln \cos \phi_{1/2}}$, with $\phi_{1/2}$ corresponding to the

semi-angle at half illuminance. The concentrator gain is denoted as $g(\phi_{ef})$

$$g(\phi_{ef}) = \begin{cases} \frac{n^2}{\sin^2 \phi_{FOV}} & \text{if } 0 \leq \phi_{ef} \leq \phi_{FOV} \\ 0 & \text{if } \phi_{ef} > \phi_{FOV}, \end{cases} \quad (2)$$

where ϕ_{FOV} denotes the receiver's FOV, while n is the refractive index derived as the ratio of the speed of light in vacuum to that in the optical material.

The mathematical modeling of channel gain in NLOS conditions is described as [6]

$$G_{NLOS} = \sum_{l=0}^{\infty} G^{(l)}, \quad (3)$$

where l denotes the reflection index, and channel gain $G^{(l)}$ after the l reflection. The non-ideal channel gain between the e th LiFi AP and the f th UE is expressed as

$$G_{ef} = G_{NLOS} + G_{ef}^v \text{ for } e \in \mathcal{E} \setminus \{0\}. \quad (4)$$

3.1.2. RF channel gain

The signal transmitted by RF AP and received by UE f follows the RF signal propagation model. The signal received accounts for both fading and path loss. The received RF signal power channel gain is modeled using the WINNER-II [15] channel model. The channel gain between the UE and RF AP is modeled as

$$G_{0e} = L d_{t_{0e}}^{-a_{0e}} \chi_{0e}, \quad (5)$$

where χ_{0e} represents the Nakagami fading channel, $d_{t_{0e}}$ is the distance between UE and RF AP, and a_{0e} indicates the path-loss exponent. Here, the value of L is determined by $L = 10^{X/10}$ where $X = M + N \log_{10} \left(\frac{f_c}{5} \right)$ describes a propagation model where f_c represents the carrier frequency in GHz, and M and N are constants specific to the propagation environment. In LOS scenario, the values are $M = 46.8$ and $N = 20$, whereas in NLOS scenario, the values are $M = 43.8$ and $N = 20$. The Nakagami fading channel, denoted as χ_{0j} , follows a gamma distribution for the fading power.

3.1.3. Received RF and LiFi signals

Since our ultimate goal is the maximization of data rate for users. The total data rate maximization is characterized by achievable sum rate maximization. Achieving achievable sum rate maximization involves the study of step-by-step analysis of the data transmission and reception at the transmitter and the receiver respectively. Thus, we explain the mathematical model of the received data, which is the transmitted data multiplied by the channel gain and added with the noise at the receiver. The received data also consists of the interference signals received from the non-desired transmitters. Once the data is received at the receiver, the achievable sum rate is formulated with the help of Shannon capacity formula. This formula tells the data rate on an individual transmitter-receiver link. For the LiFi AP, intensity modulation and direct detection (IM/DD) techniques using the Shannon's capacity formula is used. The Shannon capacity formula is directly dependent on the bandwidth and signal-to-interference-plus-noise-ratio (SINR) values. The equation of SINR is the ratio of product of power and channel gain to product of noise and bandwidth. Since we are using multiple LiFi APs, consideration of interfering APs is done in the denominator. Finally the throughput maximization equation is formed with help of this. Therefore, in this section we have tried to explain the strategy for achievable data rate maximization and communication channel modeling. Here, we will try to explain it in a more detailed version. When the UE f communication is established from an RF AP $e = 0$, the signal received by a UE f is given as [11]

$$v_e = \sqrt{G_{0e} P_0} \times x_0 + n_f^f, \quad (6)$$

where n_f^f represents additive white Gaussian noise (AWGN), however, the scenario changes when UE f is establishing communication with LiFi APs. The signal received is given as

$$v_e = \rho_f G_{ef} P_e x_e + \sum_{k \in \mathcal{E} \setminus \{e\}} \rho_f G_{kf} P_k x_k D_k(\alpha_{kf'}) + n_f^v, \quad (7)$$

where ρ_f represents responsivity, and n_f^v is thermal and shot noise. Additionally, for idle APs, the term $D_k(\alpha_{kf'})$ is expressed as $D_k(\alpha_{kf'}) = (1 - \prod_{f' \in \mathcal{F} \setminus \{f\}} (1 - \alpha_{kf'}))$. Here, $\alpha_{kf'}$ is an indicator function indicating the association of UE f' with AP k , given as

$$\alpha_{kf'} = \begin{cases} 1 & \text{if UE } f' \text{ is associated to AP } k \\ 0 & \text{otherwise.} \end{cases} \quad (8)$$

After illustrating the parameters used in our work, we formulated the Shannon capacity based achievable sum rate expression.

3.2. Shannon capacity and achievable sum rate

LiFi communication uses IM/DD of light signals. The authors in [16] have approximated the channel capacity (CC) of IM/DD using Shannon's capacity formula as a lower bound

$$CC = \frac{1}{2} B \log_2 \left(1 + w \frac{\rho^2 P_t^2}{\sigma^2} \right), \quad (9)$$

where w is a constant defined as $e/2\pi$ (with e being Euler's number), B denotes modulation bandwidth, P_t signifies optical power received, and σ^2 stands for Gaussian noise power. The factor of $1/2$ arises due to different constraints. Hence, achievable data rate is given as

$$r_{ef} = \begin{cases} B_{0f} \log_2 (1 + \text{SINR}_{0f}), & \text{for } e = 0 \text{ and} \\ \frac{1}{2} B_{ef} \log_2 (1 + w \text{SINR}_{ef}), & \text{for } e \in \mathcal{E} \setminus \{0\}, \end{cases} \quad (10)$$

where SINR_{0f} and SINR_{ef} is the signal-to-interference-plus-noise ratio serve as the lower and upper bounds, respectively, and are provided as follows

$$\begin{aligned} \text{SINR}_{0f} &= \frac{P_0 G_{0f}}{n_f^f B_{0f}}, \text{ and} \\ \text{SINR}_{ef} &= \frac{G_{ef}^2 P_e^2}{n_f^f B_{ef} + \sum_{k \in \mathcal{E} \setminus \{f\}} \rho_e G_{kf}^2 P_k^2 (1 - \prod_{f' \in \mathcal{F} \setminus \{f\}} (1 - \alpha_{kf'}))}, \end{aligned} \quad (11)$$

where the bandwidths of the RF AP-UE pair and LiFi AP-UE pair are denoted as B_{0f} and B_{ef} , respectively. With this information, the throughput of the AP can be determined as

$$r_e = \sum_{f \in \mathcal{F}} \alpha_{ef} r_{ef}. \quad (12)$$

The achievable sum rate will be given as

$$r = \sum_{e \in \mathcal{E}} r_e \quad (13)$$

The achievable sum rate represents the throughput of the hybrid RF/LiFi system. Higher the achievable sum rate, better will be the quality of service (QoS) given to the users.

3.3. Problem addressed

We address the problem of maximizing the achievable sum rate in an indoor hybrid RF/LiFi system, obtained in (13). This maximization is subjected to constraints on allocation strategies, bandwidth and power. As the allocation strategy involves an indicator parameter, this problem becomes a non-convex integer optimization problem. Conventional optimization algorithms to solve this problem are based on presumption of one of the optimization parameters. However, as every optimization parameter impacts the other, such presumptions can lead to suboptimal results. We propose the application of deep learning methods to address this issue. Thus, the objective of our work is to solve this optimization problem with the help of DDPG algorithm. The optimization goal is explained in detail in the next section.

4. Optimization goal

The optimization goal is to maximize the achievable sum rate at the UE while adhering to all the constraints with balancing the load. The DRL algorithm associates the UE to an AP which satisfies the constraints and assigns the optimal power and bandwidth in the system. The action vector represents the value for the transmission power, bandwidth and association parameters. The state vector contains the observation state and constraints that must be fulfilled to make the reliable connection between a UE and an AP. After observing the state of the environment, DRL agent chooses an action which maximizes the reward by satisfying all the constraints.

The fundamental objective of hybrid RF/LiFi is to maximize the total achievable sum rate. However, this goal is subject to constraints such as bandwidth, power, and SINR. We assume that LiFi APs have maximum available bandwidth $B_{\max}^{\text{LiFi}}$ and maximum allowable transmission power $P_{\max}^{\text{LiFi}}$, while RF APs have similar parameters $B_{\max}^{\text{RF}}$ and $P_{\max}^{\text{RF}}$. Additionally, let y_{ef} represent the minimum SINR threshold for a $e - f$ pair. Denoting z as $[B_{ef}, P_e, \alpha_{ef}]$, the resource allocation problem is formulated as follows

$$P : \max_{B_{ef}, P_e, \alpha_{ef}} r_e, \quad \text{for } e \in |\mathcal{E}|, f \in |\mathcal{F}|, \quad (14)$$

subject to

$$C_1 : \sum_{f \in \mathcal{F}} \alpha_{ef} B_{ef} \leq B_{\max}^{\text{LiFi}}, \quad \text{for } e \in \mathcal{E} \setminus \{0\}, \quad (15)$$

$$C_2 : \sum_{f \in \mathcal{F}} \alpha_{ef} B_{ef} \leq B_{\max}^{\text{RF}}, \quad \text{for } e = 0, \quad (16)$$

$$C_3 : 0 \leq P_e \leq P_{\max}^{\text{LiFi}}, \quad \text{for } e \in \mathcal{E} \setminus \{0\}, \quad (17)$$

$$C_4 : 0 \leq P_e \leq P_{\max}^{\text{RF}}, \quad \text{for } e = 0, \quad (18)$$

$$C_5 : \text{SINR}_{ef} \geq y_{ef}, \quad \text{for } e \in \mathcal{E}, f \in \mathcal{F}, \quad (19)$$

When equality is attained in constraint (19), the following criteria are obtained to prevent SINR violation

$$1 - \sum_{e \in \mathcal{E}} \sum_{f \in \mathcal{F}} \zeta_{ef} > 0, \quad \text{and} \quad (20)$$

$$\sum_{e \in \mathcal{E}} \sum_{f \in \mathcal{F}} \kappa_e \zeta_e \leq 1,$$

where

$$\zeta_{ef} = \left(1 + \frac{1}{y_{ef}}\right)^{-1}, \quad \text{and} \quad (21)$$

$$\kappa_{ef} = \frac{N_0 B_{ef}}{(G_{ef} P_e / y_{ef}) - N_0 B_{ef}} + 1. \quad (22)$$

There is a probability that one AP will deliver a high SINR. A large number of UEs may associate with it, reducing individual data rates. Thus, the final constraint is enforced for load balancing. If b_e is the UEs associated with an AP, then maximum number of UEs can associate is b

$$C_6 : b_e \leq b \quad (23)$$

The problem in (14) is a non-concave integer optimization problem. Conventional optimization algorithms assume one of these values. Therefore, we apply model-free DRL algorithms to solve the problem with mentioned constraints.

5. Framework

We have formed an action space which contains the transmit power, bandwidth and association parameter. The state space represents the two things, firstly it contains the information about the location and positioning of UE and AP. The second part ensures that all the constraints mentioned from Eqs. (15) to (23) are satisfied. The value 0 to 1 tell how strongly it is satisfied. If the constraints are not satisfied, the DRL agent comes back to the CU and the association will not happen. When all the constraints are satisfied, the reward, i.e., achievable sum-rate is maximized for an AP-UE pair.

5.1. Action space

The multiple agents in the system take decisions based on the present state of the environment. Since we are using DDPG therefore, we are considering continuous multi-dimensional action space. Each variable in the action space involves three parameters, transmission power (P_e), AP-UE link bandwidth (B_{ef}), and association parameter (α_{ef}), $1 \leq i \leq l$ where l is the size of action space for each e th LiFi AP - f th UE link. Allocating downlink bandwidth and transmission power to the APs has a substantial impact on the system's possible sum-rate, in addition to determining the affiliation of UEs with APs for downlink data reception. DDPG's continuous action space provides greater precision compared to DQN's discrete action spaces. Each agent's action (RF and LiFi APs) represents the transmit power, bandwidth and association parameter vector for each user in the system. If the user is not in LOS with the LiFi AP, their transmit power and bandwidth is 0, and its gets associated to RF AP. The RF AP uses the actor-network's output to perform actions.

The allocation strategy association parameter (α_{ef}): We first include the allocation strategy into the action space. In this regard, the first parameter included in the action space is the association parameter α_{ef} . The association parameter is an indicator variable which tells the association of an AP to a UE. The association is based on the level of SINR received. DRL agent works such that the UE gets associated to an AP which provides the higher SINR in its FOV with load balancing.

Power (P_e): The third parameter included in the action space is the power (P_e). As our aim is not only to maximize the achievable sum-rate but also consider eye safety and power saving. As we have put up a constraint on maximum power that a control unit (CU) can allocate to an AP, hence among the values of action vector, highest value is chosen from the action vector to maximize the achievable sum-rate.

Bandwidth (B_{ef}): Once the association parameter has been formulated, the next issue is to find strategies for adjusting their bandwidths. Thus, the second parameter included in the action space is the bandwidth (B_{ef}). It is known that channel capacity is directly proportional to the bandwidth. The bandwidth allocated to RF AP is given as (B_{0f}) where as for the LiFi AP it is given as (B_{ef}) such that $e \in \mathcal{E} \setminus \{0\}$. Since the available spectrum is limited therefore, we have applied the constraint on bandwidth usage. Similarly, like the allocation of transmit power, the CU allocates the bandwidth to the AP by keeping the constraint satisfied. The AP provides the association to UEs while balancing the load. Throughout this process the CU monitors the action so that the value chosen does not exceed the maximum limit in order to achieve the higher data rate.

The action vector for e LiFi APs and an RF AP are given as

$$\mathcal{A} = \left\{ (P_{e_1}, B_{ef_1}, \alpha_{ef_1}), (P_{e_2}, B_{ef_2}, \alpha_{ef_2}), \dots, (P_{e_l}, B_{ef_l}, \alpha_{ef_l}) \right\} \quad (24)$$

$$= \{a_{ef_1}, a_{ef_2}, \dots, a_{ef_l}\},$$

where P_e and B_{ef} are power and bandwidth allocations, respectively, for the e th AP and f th UE link. α_{ef} denoted the association parameter for UE e and AP f link. These values are scaled to match operational parameters within the environment, with P_{e_i} constrained within $[P_{\min}, P_{\max}]$ and B_{ef_i} within $[B_{\min}, B_{\max}]$, where $P_{\min}, B_{\min}$ and $P_{\max}, B_{\max}$ are the minimum and maximum permissible levels for power and bandwidth, respectively.

5.2. State space

In the proposed hybrid RF/LiFi system, the state space s_f provides the observation of the agent (CU) in the environment, which includes

1. The locations of the UEs and APs at an instantiation, represented by $x_f, y_f, \text{AP}_{x_f}$ and AP_{y_f} .

- The outcome of the action of the agent (CU) observed on the environment. The action of CU is choosing a bandwidth, power and association parameter. The outcome of this action is observed on the constraints imposed on the system, represented by $C_1, C_2, C_3, C_4, C_5, C_6$.

In [10,11], authors have used state space as a constraint. For the environment's state vector, we employ the constraints along with UEs-APs location at any given moment. The state s can be described as $[x_f, y_f, AP_{x_f}, AP_{y_f}, C_1, C_2, C_3, C_4, C_5, C_6]$, where the former part x_f, y_f are the normalized coordinates of UEs f 's location and AP_{x_f}, AP_{y_f} are the normalized location of the AP e serving UE f . The latter part are the constraints mentioned in (15) to (23). The rationale for choosing the constraints in the state space is that it directly affects the optimal resource allocation and data rate. Hence, the state for f users is defined as $s = [s_1, s_2, \dots, s_f]$. Each user state s_f is given as

$$s_f = [x_f, y_f, AP_{x_f}, AP_{y_f}, C_1, C_2, C_3, C_4, C_5, C_6]. \quad (25)$$

Note that C_m takes on values in the set 0,1 for all $m = 1, 2, \dots, 6$, with a value closer to 1 indicating the degree to which it effectively satisfies the corresponding constraint. C_m becomes zero when any of the constraint is not satisfied.

5.3. Reward

The reward function aim is to optimize network performance and maximize overall reward. It is mathematically expressed as

$$R(s, a) = r = \sum_{f \in \mathcal{F}} \alpha_{ef} r_{ef}. \quad (26)$$

To use DDPG, we must define the problem in continuous state space, action space and rewards as mentioned. In DDPG, neural networks are trained using the set (s_t, a_t, r_t, s_{t+1}) .

5.4. Complexity analysis

Each agent's Q-table has $3F$ rows reflecting environment state and $|P|^F$ columns representing actions. The Q-Table has $3F \times |P|^F$ entries, indicating exponential growth as the number of UEs increases. Each DDPG agent (i.e., AP) has four neural networks, including actor, critic, and target networks for improved stability. The actor and critic networks have two hidden layers, each with $2F$ nodes. Actor networks have $2F$ inputs and F outputs (transmit power, bandwidth and association parameter for each UE). The DDPG-based method scales to accommodate additional UEs while growing complexity. DDPG outperforms DQN in terms of convergence performance as the number of UEs F increases. This is because DDPG's network expands linearly, whereas DQN grow exponentially.

6. Simulation results

We considered the system model as shown in Fig. 1. To assess its practicality, we utilized a Gymnasium environment developed in Python. Simulation data are taken from [17]. The setup includes 8 LiFi APs surrounding the RF AP arranged on the room's ceiling. The area of the room considered is 10 m × 10 m with specific parameters mentioned in Table 1.

6.1. Performance order

We compared our proposed algorithm DDPG with PPO, TD3, DQN, and DDQN on several performance metrics. We initially opted for DQN because DDPG efficiently handles continuous action spaces, unlike the DQN-based learning algorithm which is suited for discrete actions. The evaluation is based on detailed simulations running over 2000 episodes within a hybrid RF/LiFi system with learning rates 0.0003.

Algorithm 1 DDPG for resource allocation in hybrid RF/LiFi System

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1: Initialize critic network  $Q(s, a|\theta^Q)$  and actor  $\mu(s|\theta^\mu)$  with weights  $\theta^Q$  and  $\theta^\mu$ 
2: Initialize target network  $Q'$  and  $\mu'$  with weights  $\theta^{Q'} \leftarrow \theta^Q, \theta^{\mu'} \leftarrow \theta^\mu$ 
3: Initialize replay buffer  $D$ 
4: for episode = 1, M do
5:   Initialize a random process  $N$  for action exploration
6:   Receive initial observation state  $s_1$ 
7:   for t = 1, T do
8:     Select action  $a_t = \mu(s_t|\theta^\mu) + N_t$  according to the current policy and exploration noise
9:     Execute action  $a_t$  and observe reward  $r_t$  and new state  $s_{t+1}$ 
10:    Store transition  $(s_t, a_t, r_t, s_{t+1})$  in  $D$ 
11:    Sample a random minibatch of  $N$  transitions  $(s_i, a_i, r_i, s_{i+1})$  from  $D$ 
12:    Set  $y_i = r_i + \gamma Q'(s_{i+1}, \mu'(s_{i+1}|\theta^{\mu'}))|\theta^{Q'}$ 
13:    Update critic by minimizing the loss:
        
$$L = \frac{1}{N} \sum_i (y_i - Q(s_i, a_i|\theta^Q))^2$$

14:     $\theta^Q \leftarrow \theta^Q - \alpha \nabla_{\theta^Q} L$ 
15:    Update actor policy using the sampled policy gradient:
        
$$\nabla_{\theta^\mu} J \approx \frac{1}{N} \sum_i \nabla_a Q(s, a|\theta^Q)|_{s=s_i, a=\mu(s_i)} \nabla_{\theta^\mu} \mu(s|\theta^\mu)|_{s_i}$$

16:    Update the target networks:
        
$$\theta^{Q'} \leftarrow \tau \theta^Q + (1 - \tau) \theta^{Q'}$$

17:   end for
18: end for

```

Table 1
Simulation parameters.

Li-Fi parameter	Values
Number of RF AP	1
Number of LiFi APs	8
Height of room	5 m
Number of UEs	8
Area of photodiode	1 cm ²
LiFi AP average optical power	9.2 W
Noise at LiFi AP	10 ⁻²¹ A ² /Hz
Responsivity	0.28 A/W
FOV at UE	60°
Users speed	0 to 0.5 m/s randomly
Area size	10 × 10 m ²
Bandwidth	20 MHz
Semi-angle of LiFi APs	70°
Hyper-parameter	Values
Number of episode	2000
Learning rate	0.0003
Discount factor	0.9
DDPG architecture	Activation
Fully connected actor layer	Sigmoid
Fully connected critic layer	ReLU

It outperforms the DQN in all the scenarios. To mark the robustness of our proposed algorithm, we compared it with several other mature algorithm like DDQN, TD3 and PPO. Remarkably, DDPG outperforms all of them, i.e., PPO, TD3, DQN, and DDQN in all the metrics.

Fig. 2a compares the performance of DDPG, PPO, TD3, DQN, and DDQN in terms of the achievable sum rate (Mbps) over the number of episodes. A higher value typically indicates better performance.

DDPG shows steady improvement over the episodes, with the sum rate eventually stabilizing around 1750 Mbps. This indicates that DDPG successfully learns to optimize the sum rate efficiently over time. PPO

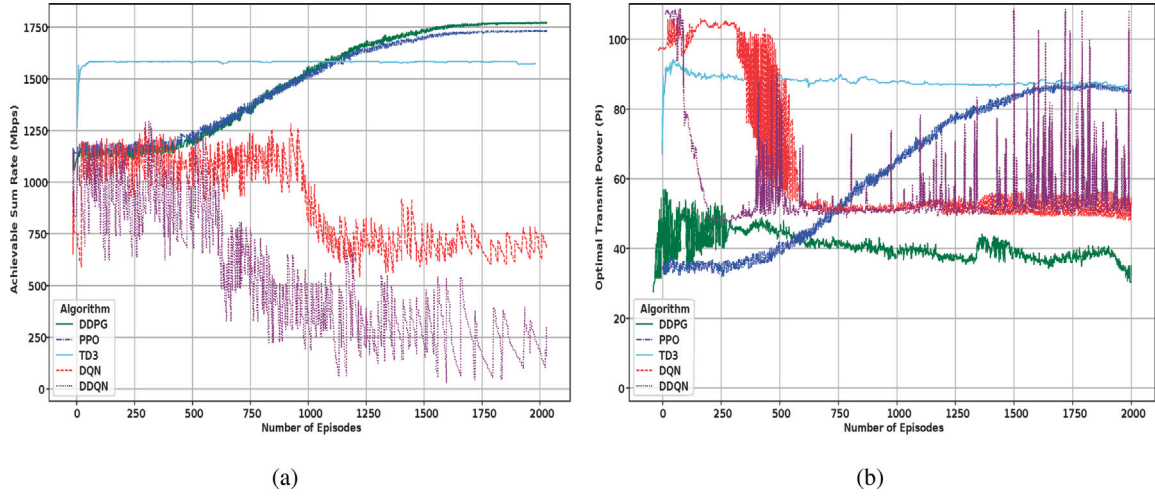


Fig. 2. (a) Comparison of DDPG with DQN, DDQN, PPO, and TD3 based learning algorithms at learning rate 0.0003. (b) Convergence of DRL algorithms in terms of mean power consumption at learning rate 0.0003.

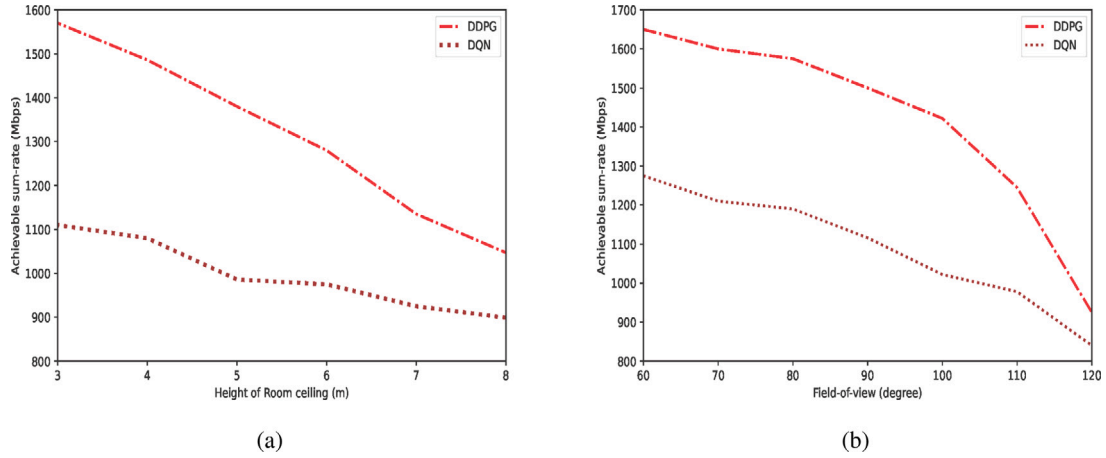


Fig. 3. (a) Convergence with height (b) Convergence with field of view at learning rate 0.0003.

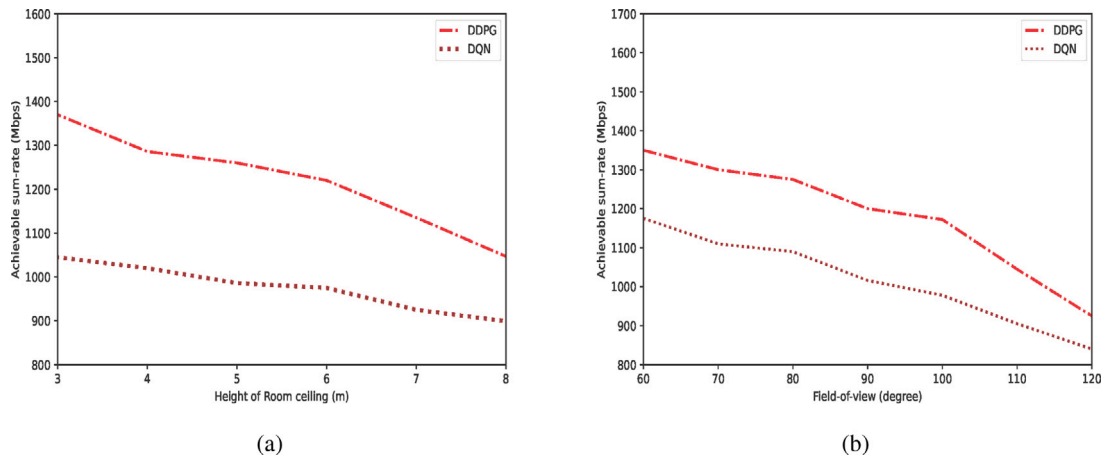


Fig. 4. (a) Convergence with height (b) Convergence with field of view at learning rate 0.03.

demonstrates similar steady growth as DDPG, eventually reaching a slightly lower sum rate, stabilizing close to 1700 Mbps. This suggests good performance in optimizing the sum rate but slightly less efficient compared to DDPG possibly because in low noise conditions with slow movements of UEs, DDPG performs better. TD3, this variant of DDPG performs well but not as comparable to DDPG. TD3, although also an

actor-critic method, employs twin critics and target policy smoothing to reduce overestimation bias, but in our set-up, there is relatively low-dimensional state spaces and action spaces, where the complexity is not much high and also low noise conditions. Intuitively, we can say that is why DDPG's simplicity makes it to perform better in our set-up.

This indicates that TD3 is less effective in this environment. DQN's performance is highly volatile with frequent fluctuations, showing a lack of stability and inconsistent improvement. DDQN similar to DQN, also exhibits significant volatility and fails to consistently improve the sum rate. DDPG and PPO shows stable learning and convergence to higher sum rates compared to DQN and DDQN. This is likely due to their ability to handle continuous action spaces more effectively. Among the algorithms, DDPG and PPO achieve the best results, indicating their suitability for this kind of set-up. TD3 also performs well, though slightly lower than DDPG and PPO. DQN and DDQN, being better suited for discrete action spaces, struggle in this environment.

Fig. 2b compares the performance in terms of the optimal transmit power (P_t) over the number of episodes. For DDPG, the transmit power fluctuates initially but stabilizes around 35–40 mW after about 500 episodes. This suggests that DDPG is relatively stable for optimal transmit power, optimizing it for best possible way in our set-up. For PPO the transmit power starts lower, around 35 mW, but reaches higher transmit power as the learning proceeds as compared to DDPG. TD3 quickly stabilizes around 70–80 mW of transmit power. TD3's stable performance but higher power usage suggest it requires higher transmission power. DQN displays significant volatility in transmit power, starting around 90–100 mW. DQN and DDQN struggles to find a stable policy for controlling power in this scenario. This instability indicates difficulties in optimizing transmit power possibly due to their inability to handle continuous action and state spaces. DDPG, PPO, and TD3 show more stable convergence to a particular power level, indicating better handling of the continuous action space and the associated optimization problem. DQN and DDQN, which are typically better for discrete actions, struggle to stabilize, highlighting their limitations in continuous domains. It indicates that algorithms designed for continuous action spaces (like DDPG, PPO and TD3) are better suited for tasks involving transmit power control in hybrid RF/LiFi systems.

Figs. 3(a) and 4(a) depicts the achievable sum-rate with varying height of the ceiling in room but with different learning rates, demonstrating the expected reduction in sum-rate with increasing height. Figs. 3(b) and 4(b) shows the achievable sum-rate for different values of FOV. As expected, increasing the FOV reduces overall achievable sum-rate. In all the cases, DDPG performs better than DQN.

6.2. Discussion on comparison with DQN, DDQN, TD3 and PPO

We have compared our proposed DDPG algorithm with PPO, TD3, DDPG and DQN algorithms as well. The reason for opting DDPG specifically is that it efficiently represents the continuous action space version of DQN based learning algorithm. The primary concern in our work is the increased size of action space but not extremely large in a hybrid RF/LiFi set-up considered, which also reduces the discretization in its samples. Thus, we assess a counterpart of DQN based learning (DDPG) which can efficiently handle continuous action spaces. DDPG outperforms DQN and DDQN primarily due to its ability to handle continuous action spaces effectively, which is critical in complex dynamic environments such as hybrid RF/LiFi networks. Unlike DQN and DDQN, which is limited to discrete action spaces, DDPG uses an actor-critic approach that leverages deterministic policy gradients, enabling it to optimize policies over continuous action spaces. This allows for more precise and efficient resource allocation, enhancing overall network performance.

Additionally, DDPG benefits from an off-policy training method, where the policy used to generate actions (behavior policy) can differ from the policy being improved (target policy). This separation allows DDPG to utilize past experiences more effectively, leading to better learning stability and faster convergence compared to on-policy methods like PPO. TD3, although also an actor-critic method, employs twin critics and target policy smoothing to reduce overestimation bias, but in our set-up, there is relatively low-dimensional state spaces and action spaces, where the complexity is not much high and also low

noise conditions. In low noise conditions, DDPG performs better than TD3 and PPO. Therefore, in our set-up DDPG's simplicity makes it to perform better. Several other reasons exist for the better performance of DDPG observed in this work, which are enlisted as follows:

Considering the System set-up: In our system model, we have considered static/dynamic hybrid RF/LiFi environment, i.e., UEs are either static or moving with small velocity ranging from 0 to 0.5 m/s. With this slow movement of UEs, low noise is expected to occur. DDPG performs better in low noise because of its deterministic nature. Also, due to slow dynamicity of UEs, the new data collection is slow and hence the action space is changing slowly. Unlike PPO, DDPG uses replay buffer so it can use past experiences which provides a better balance between exploration and exploitation strategy. Intuitively, we can conclude that, this is why DDPG is outperforming PPO in our system set-up.

Sample Efficiency: Being an off-policy algorithm, DDPG can reuse past experiences stored in a replay buffer, which can lead to higher sample efficiency. However, PPO an on-policy algorithm, cannot reuse past experiences to the same extent. So, this makes DDPG more sample efficient than PPO.

Handling of Continuous Action Spaces: DDPG is specifically designed for continuous action spaces, making it highly effective in environments where actions are not discrete while PPO can handle continuous action spaces, but it does not always perform as well as DDPG in these settings. It can achieve smooth and precise control in continuous action spaces. The deterministic policy of DDPG enables it to output actions within a continuous range of values, allowing for fine-grained control.

Exploration and Exploitation: DDPG utilizes a separate target network and a replay buffer which helps stabilize training and can achieve a better balance between exploration and exploitation. PPO relies on stochastic policies for exploration, which can sometimes make it harder to fine-tune the balance between exploration and exploitation.

Deterministic vs Stochastic policy: DDPG uses a deterministic policy, which simplifies the training process compared to PPO's stochastic policy. The deterministic policy allows DDPG to directly apply the current policy to old data samples from the replay buffer.

7. Comparison with age of information aware radio resource management in vehicular networks: A proactive deep reinforcement learning perspective

In [12], the authors address AoI-aware radio resource management for long-term optimization in a Manhattan grid vehicle-to-vehicle network. The roadside unit (RSU) allocates frequency bands and schedules packet transmissions for all vehicle user equipment (VUE) pairs based on the global network state at each scheduling slot. They model the stochastic decision-making process as a discrete-time, single-agent Markov decision process (MDP). They propose a proactive approach using recent advancements in long short-term memory (LSTM) and deep reinforcement learning (DRL), enabling VUE pairs to perform optimally with partial network state observations and no prior knowledge of network dynamics.

The comparison of the work in [12] with our work is as follows:

- In [12], Chen et al. have used a single agent DRL. However, in the current work, the application of DRL involves multi agent acting in different parts of the environment.
- In [12], Chen et al. study vehicle-to-vehicle (V2V) networks in an outdoor scenario. On the other hand, our system model involves the study of data rate maximization for LiFi based mobile devices in an indoor scenario.
- As the study on V2V involves vehicles in an outdoor scenario, the speed of the users is considered much more faster as compared

to our model, which involves users walking while carrying user equipments (UEs) in an indoor environment.

- As mentioned in [12], vehicular mobility results into an extremely huge state space for VUE pairs. However, in our work, the state space involves slow motion speed vectors, which result into lesser number of changes in the state of the environment, resulting into a smaller state space.
- For numerical experiments, Chen et al. have TensorFlow. However, in the current work, we have used OpenAI Gymnasium environment as it provides a more enhanced and updated simulation environment compared to TensorFlow. The basic difference between the two is that TensorFlow is a machine learning framework for building and training models while OpenAI Gymnasium is a toolkit for developing and testing RL algorithms with a variety of environments.
- In [12], Chen et al. have used DQN and LSTM, resorting to deep recurrent Q-network (DRQN). In Fig. 2, we have shown the comparison of our proposed DDPG learning based scheme with DQN based learning.

8. Conclusion

In this paper, we examined our proposed method for resource allocation in a dynamic and hybrid RF/LiFi systems. Remarkably, we used close to real scenario environment thoroughly created in Gymnasium. The proposed algorithm is compared with a few mature DRL algorithms which supports discrete and continuous action and state spaces. Simulations results shows that DDPG outperforms all of them. Also, DDPG with its capability of handling continuous action space shows 62.8% better performance than DQN in terms of achievable sum-rate and 42.8% in terms of optimal transmit power.

CRedit authorship contribution statement

Tanya Verma: Writing – original draft, Validation, Software, Resources. **Arif Raza:** Software, Formal analysis. **Shivanshu Shrivastava:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Bin Chen:** Validation, Software, Project administration, Data curation. **U.D. Dwivedi:** Resources. **Amarish Dubey:** Visualization, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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